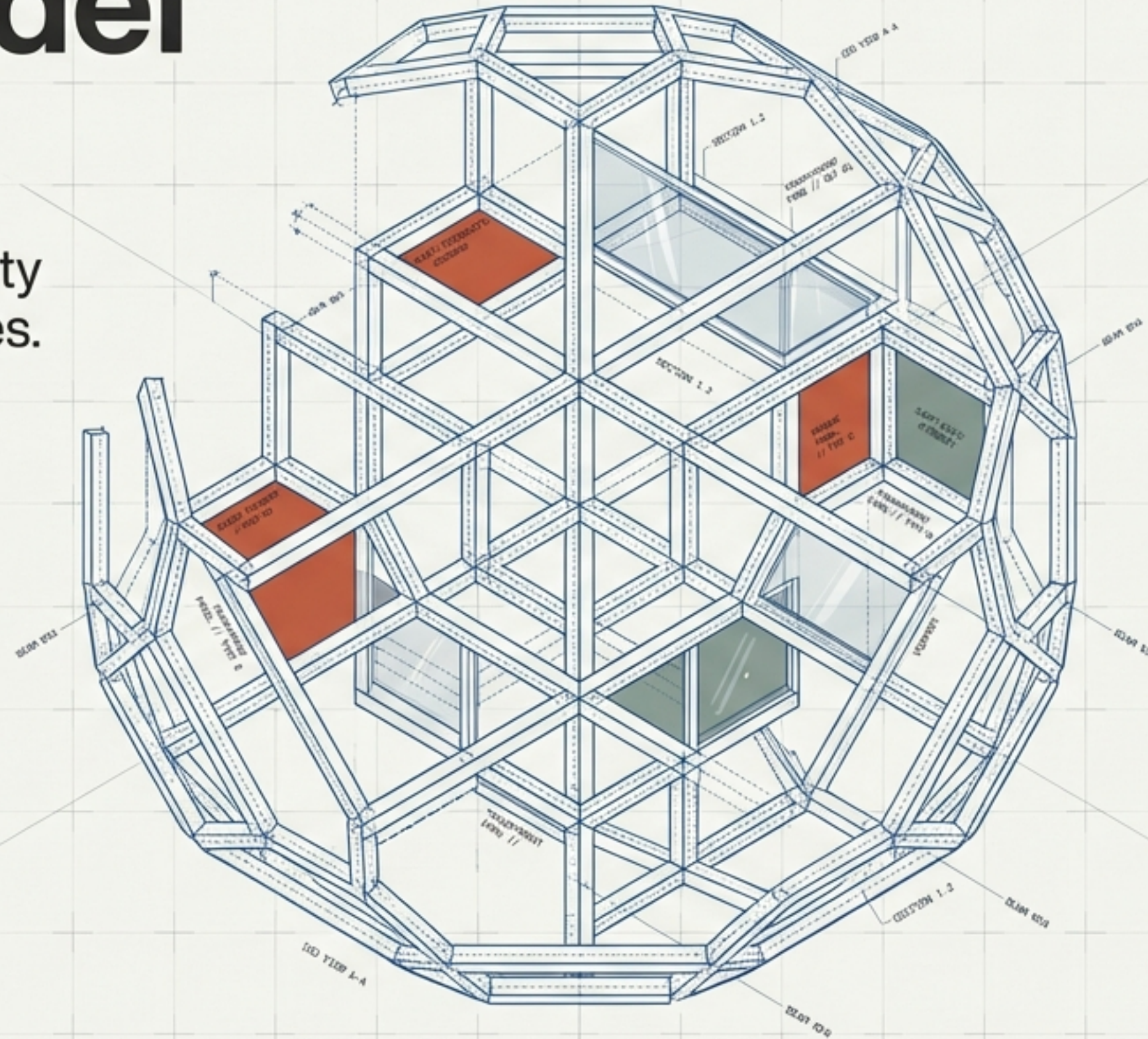
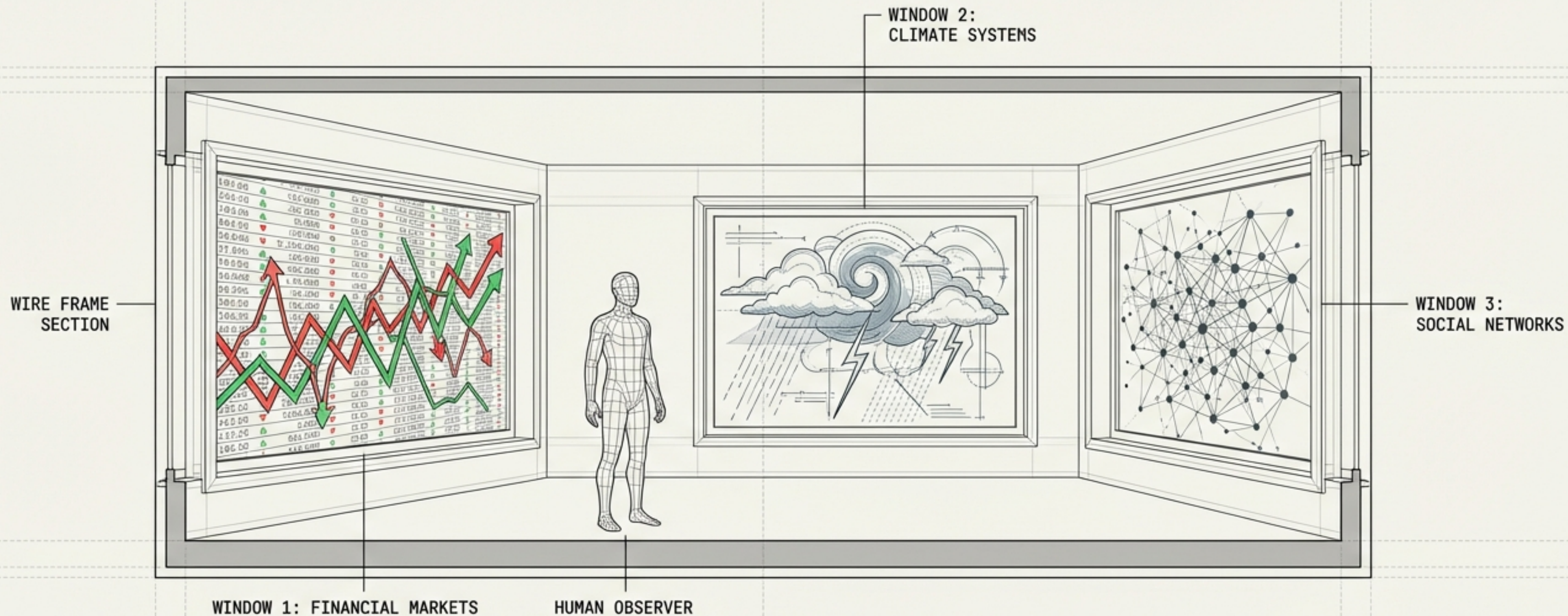


The Model Thinker

Confronting Complexity
with a Lattice of Lenses.



Georgia Pro
Based on the work of Scott E. Page



The Complexity Challenge

Our brains evolved for the savannah, optimized for simple cause-and-effect. But the modern world is a system of complex, interconnected phenomena —climate, AI, global markets.

Standing in a house with many windows allows us to triangulate the truth. Reliance on a single window is no longer a stance; it is a blind spot.

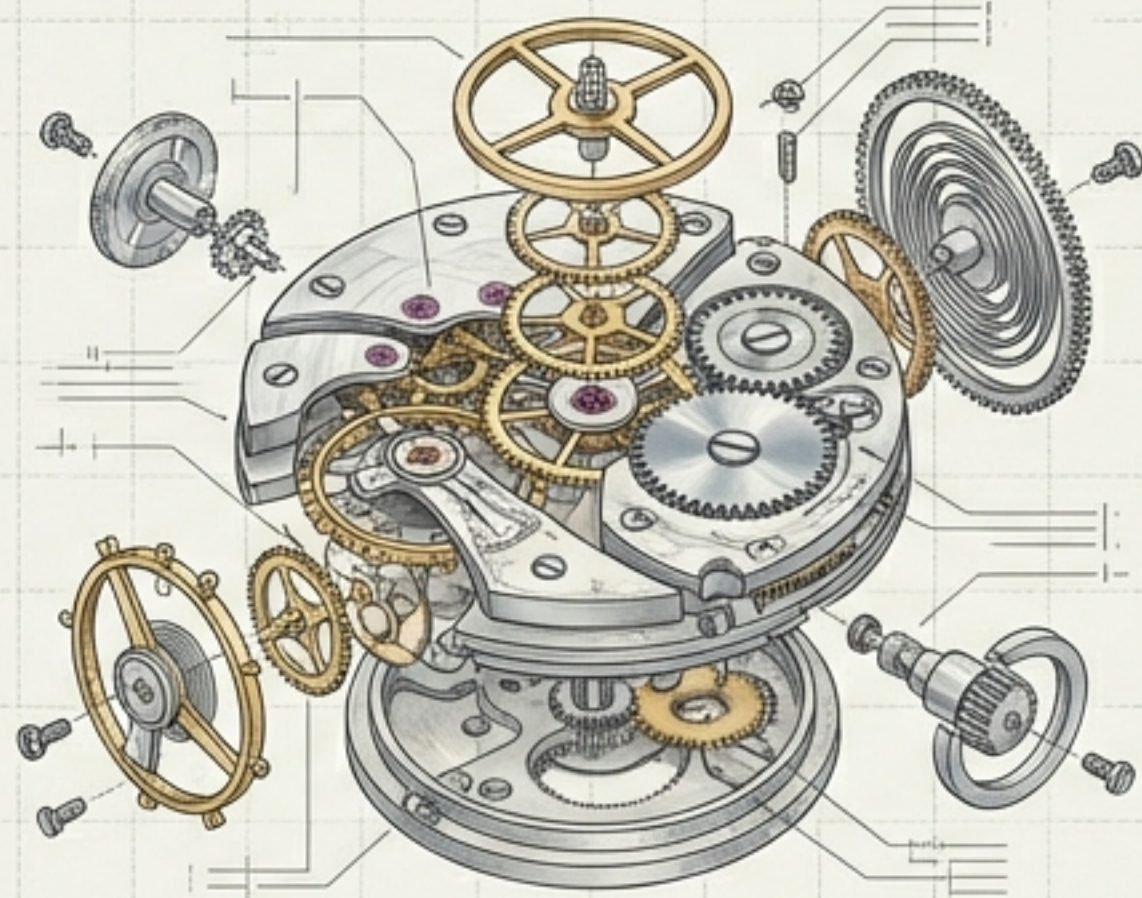
The Trade-off: Tractability vs. Realism

Ockham's Razor



Simplicity. Analytics. Solvability.

Modern Modeling



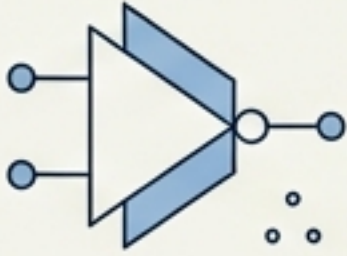
Plurality. Computation. Realism.

A model is a formal simplification of reality. Historically, we stripped away detail to make problems solvable (Ockham's Razor). Today, computational power allows us to embrace 'plurality'—boldly embrace 'plurality'—building elaborate models that sacrifice simplicity for necessary realism.

The Framework: R.E.D.C.A.P.E.

The Utility Belt of the Model Thinker

R



REASON

Identify conditions and consequences.

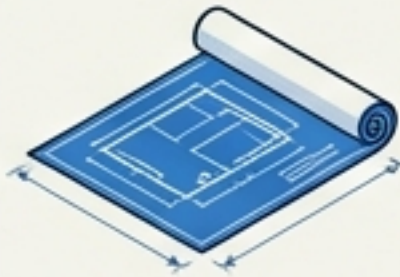
E



EXPLAIN

Provide testable mechanisms.

D



DESIGN

Build frameworks for policy. (e.g., FCC auction design)

C



COMMUNICATE

Share precise ideas.

A



ACT

Guide strategic choices.

P



PREDICT

Forecast future states. (e.g., bomb-sniffing dogs)

E

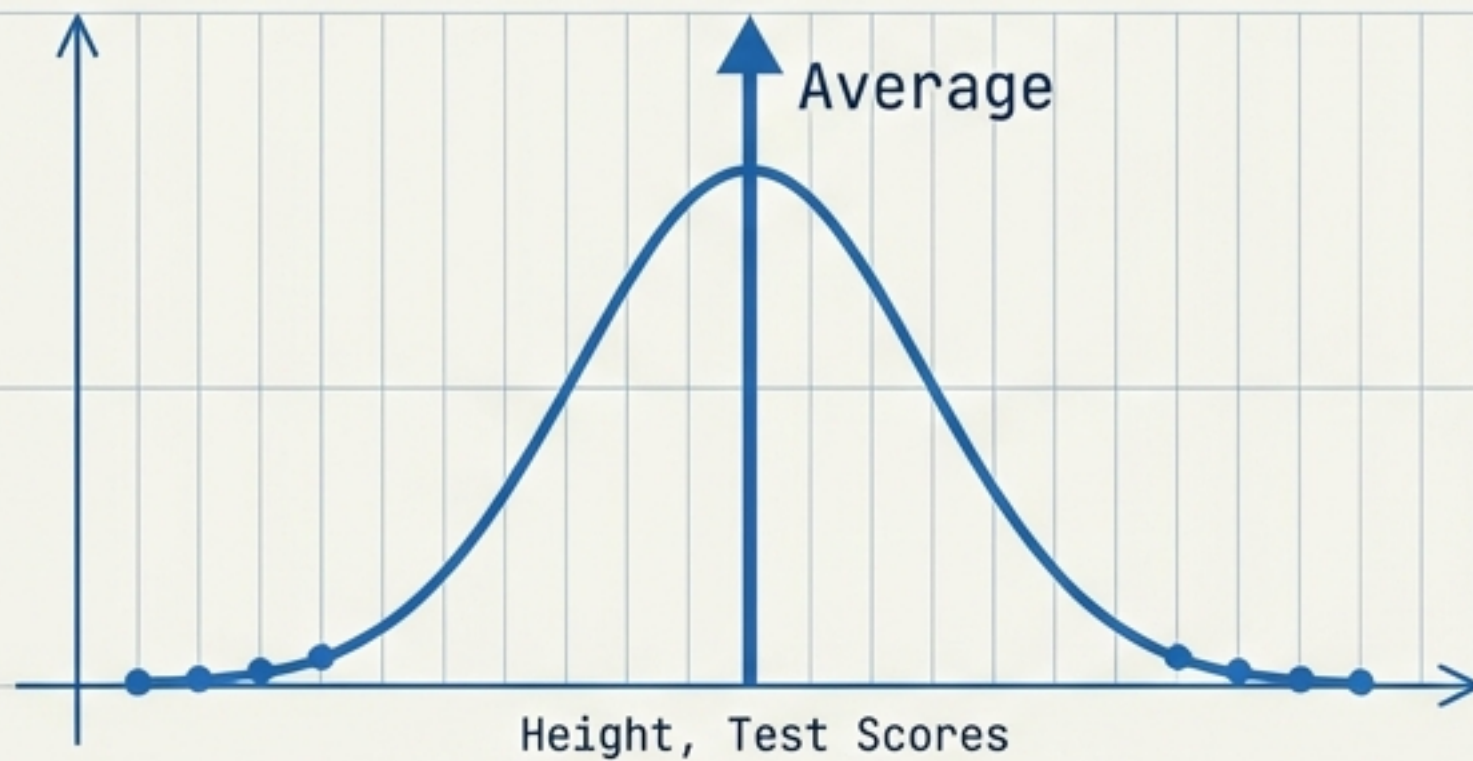


EXPLORE

Investigate 'what if' scenarios.

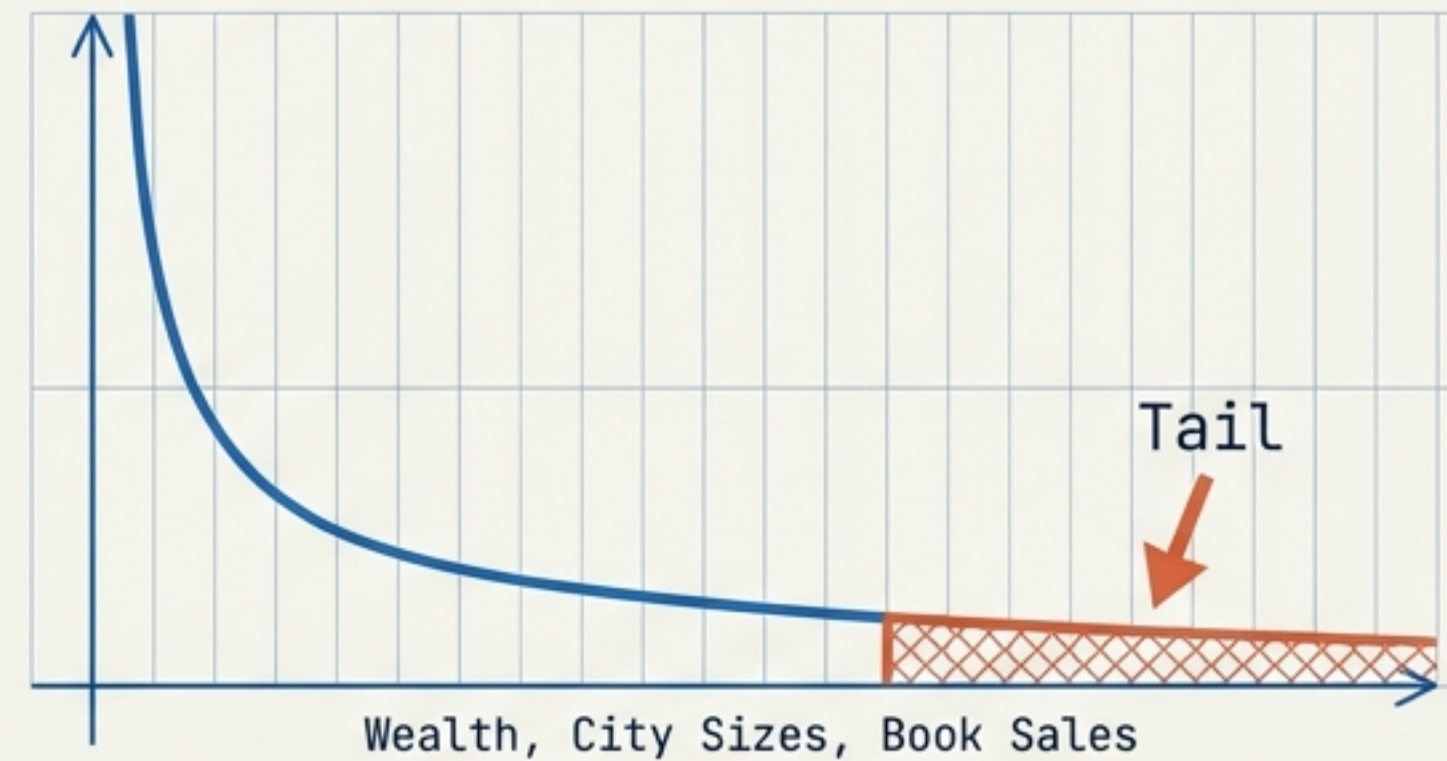
Average is Over

A. The Bell Curve (Normal Distribution)



Events cluster around the mean.
Extremes are rare.

B. The Long Tail (Power Law)

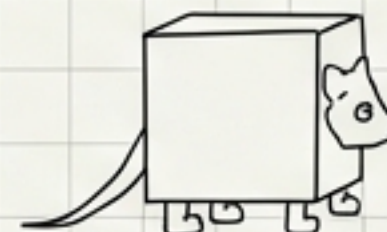
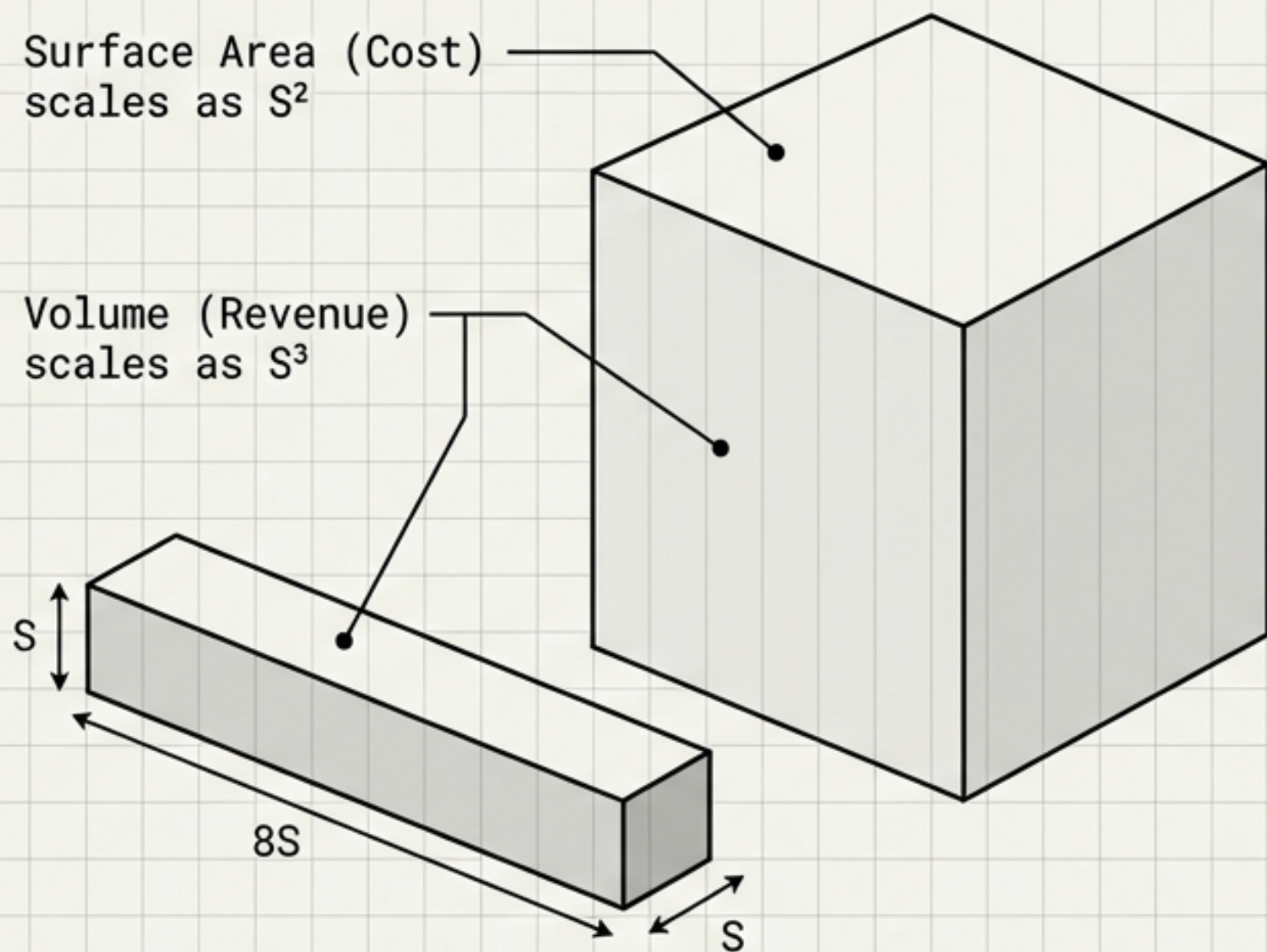


The average is meaningless.
Extremes dominate the system.

You cannot manage a Long Tail world with Bell Curve risk models.

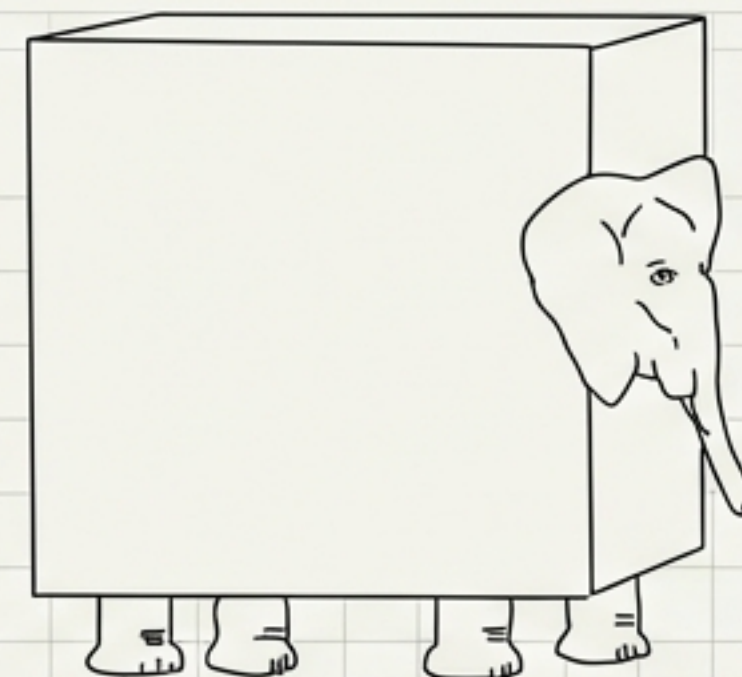
Reference: Zipf's law, Broido and Clauset (2018), book sales example: A publisher selling 5 million copies of one book earns the same revenue as selling 500 copies each of 10,000 different books. The long tail accounts for a large portion of the total value.

The Geometry of Growth



Mouse

Surface Area: 14 in^2
Volume: 3 in^3



Elephant

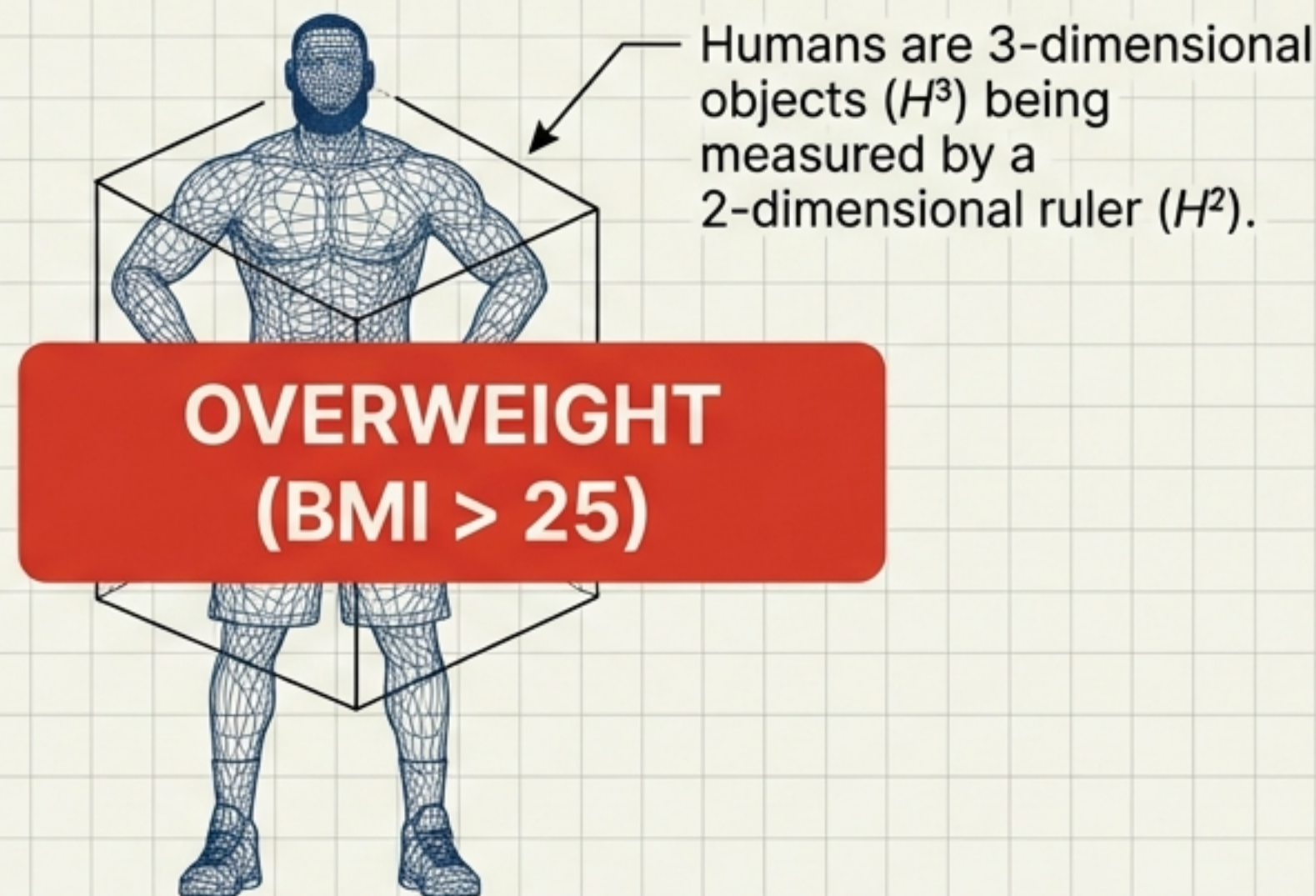
Surface Area: $57,600 \text{ in}^2$
Volume: $864,000 \text{ in}^3$

Efficiency scales non-linearly. As dimensions double, volume (revenue) increases 8x while surface area (cost) only increases 4x. This math favors bigness, explaining the existence of massive supertankers and megacities.

The Flaw of Averages: BMI

MODULE A: SCALE

$$\text{BMI} = \frac{\text{Weight}}{\text{Height}^2}$$



Scaling logic exposes the flaws in simple models. If you scale a person up, weight increases with the cube of height, but BMI only divides by the square. This misclassifies fit, tall individuals as overweight.

Exponential Growth & The Rule of 72

The Bubble



The Bubble

Rule of 72

$$\text{Years to Double} \approx \frac{72}{\text{Growth Rate}}$$

The Innovation

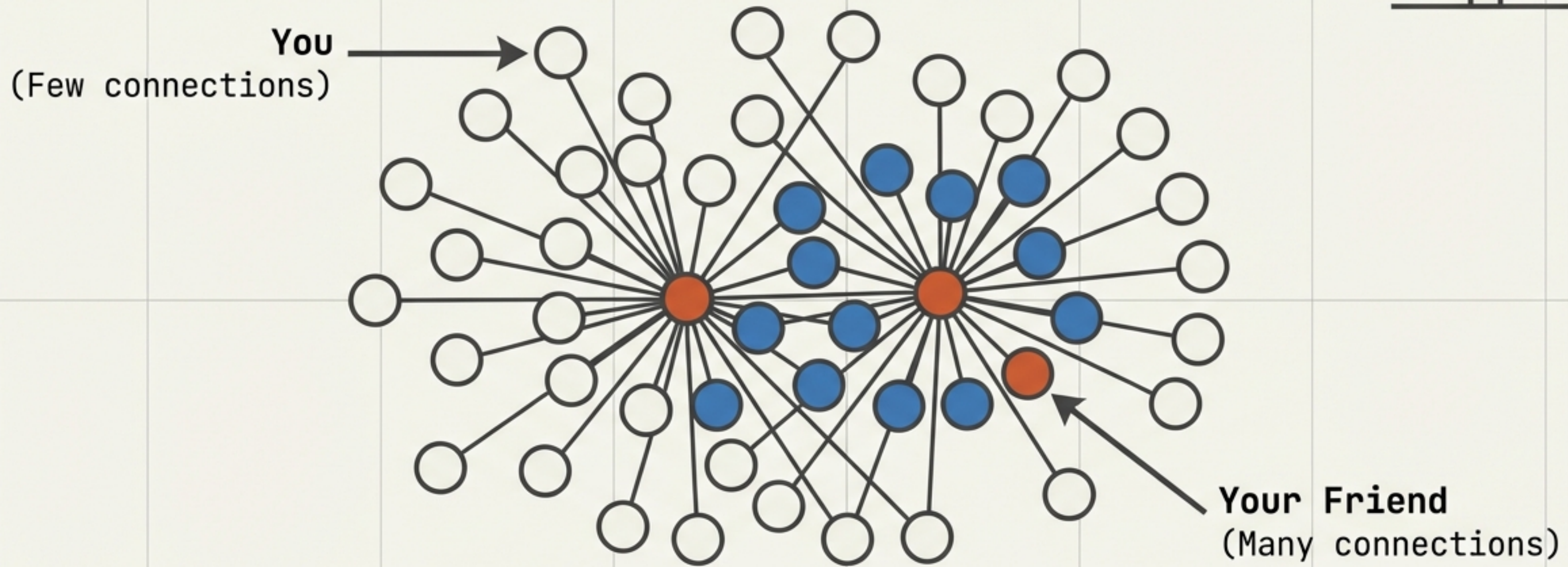


The Innovation

Growth limits depend on the mechanism. Housing prices cannot double every 7 years forever (bubble). Transistors can double every 2 years because the parameters of the system are fundamentally changed by innovation.

The Friendship Paradox

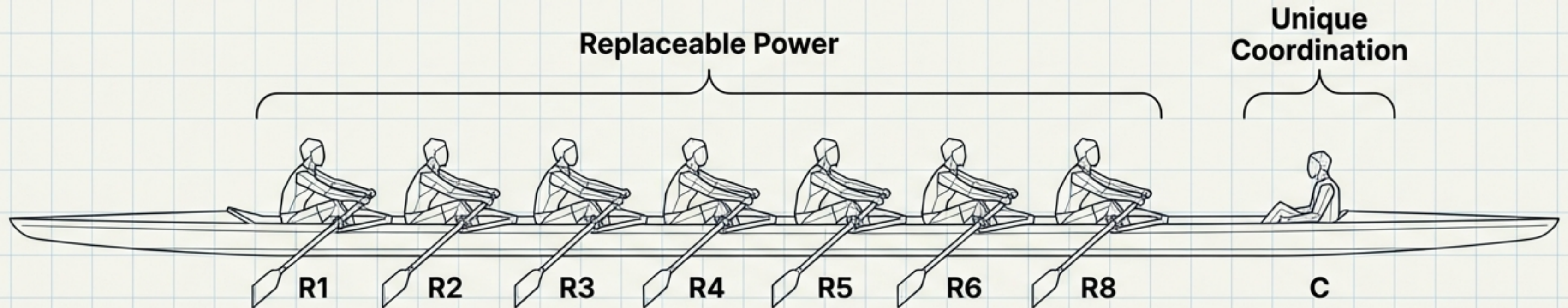
MODULE B: NETWORKS



Why do your friends have more friends than you? Because you are statistically more likely to be connected to a “Hub.” Network structure skews our perception of reality, making us feel less popular, wealthy, or successful than the average of the people we know.

Measuring Value: The Shapley Value

MODULE C: STRATEGY



Last-on-the-Bus Value: 0 for everyone (Team fails without full crew).

Shapley Value: Average marginal contribution across all arrival orders.

Result: Coxswain = 3, Rower = 1.4

Who contributes more? In a team, value isn't just output. It's the marginal contribution averaged across all possible scenarios. The Shapley Value mathematically proves the worth of the coordinator (Coxswain) even if they don't pull an oar.

Coordination Games

Matching

Pure Coordination

	Pure Coordination	
	Choice A	Choice B
Choice A	1, 1 (Success)	0, 0 (Failure)
Choice B	0, 0 (Failure)	1, 1 (Success)



Sometimes there is no 'best' option, only a 'matching' option. Whether driving on the left or right, the goal is simply to do what others do. History matters more than optimality.

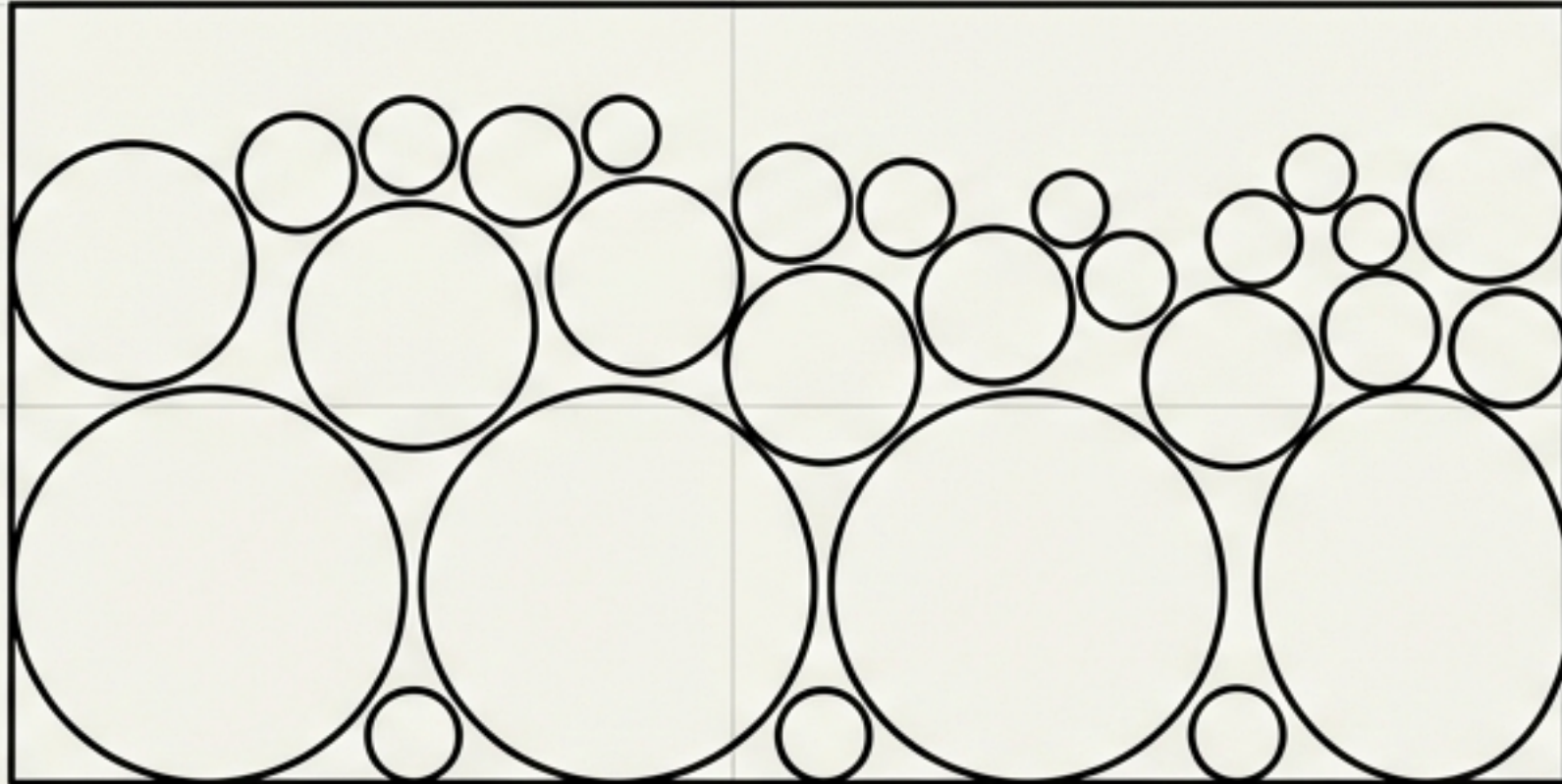
The Rational vs. The Spiteful

Absolute Gains	Relative Gains
	
Contentment (Less but okay)	Triumph (Harmed but ahead)

Do we want to do well, or do we want to beat the opponent? In 'Zero-Sum' thinking (or Neorealist politics), actors may choose to harm themselves (poke out an eye) if it means harming the rival double. This destroys value but wins the relative game.

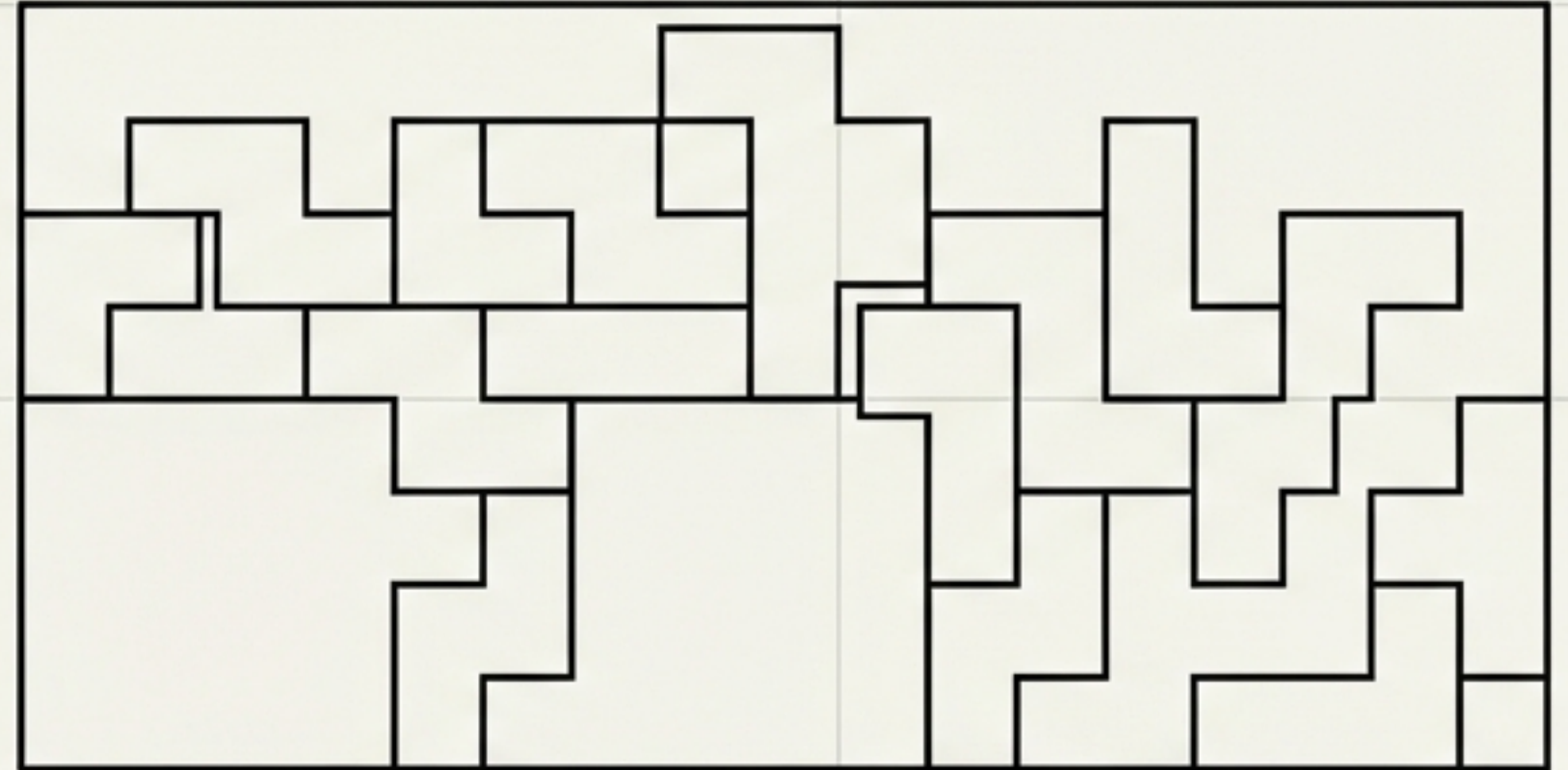
The Conditional Truth: Bin Packing

Simple Reality



Big Rocks First = Works.

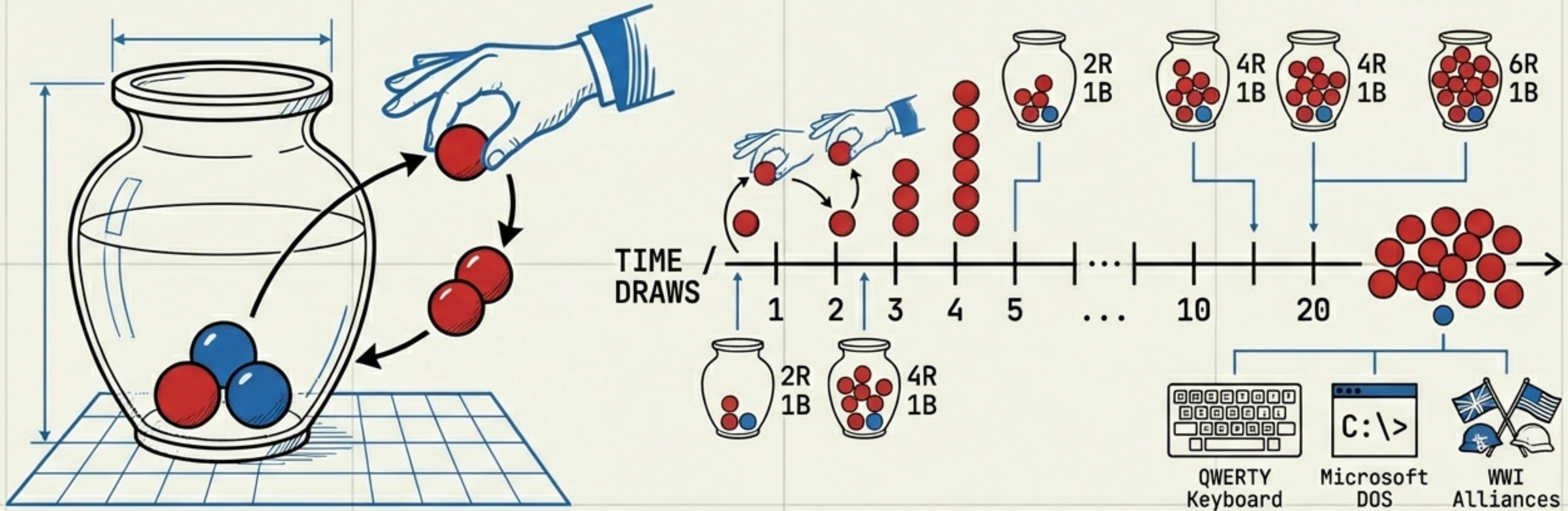
Complex Reality



Big Rocks First = Fails.

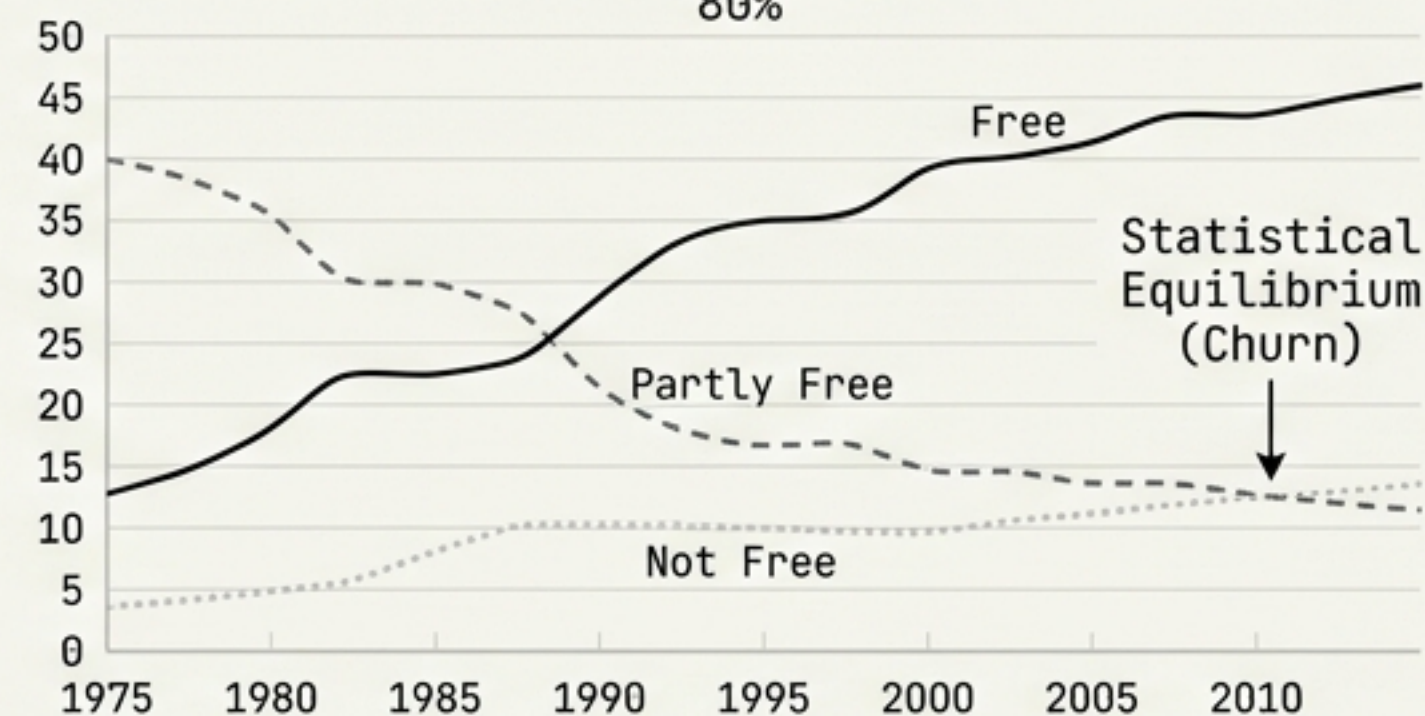
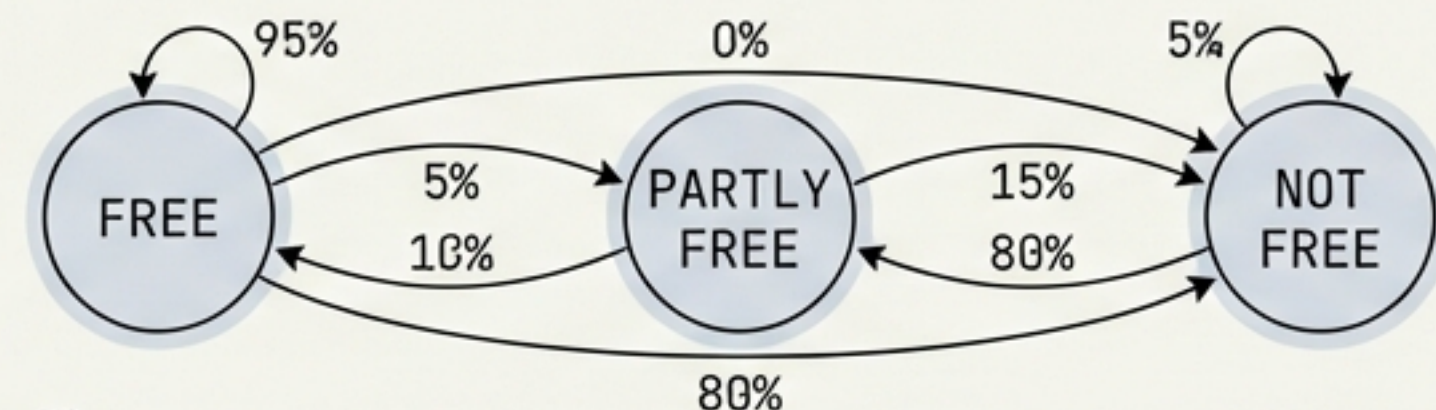
Proverbs like “Big Rocks First” are not universal truths; they are conditional. In complex, multi-dimensional problems (like packing the Space Station), intuition fails. We need algorithms to find the fit.

Path Dependence: The Polya Process



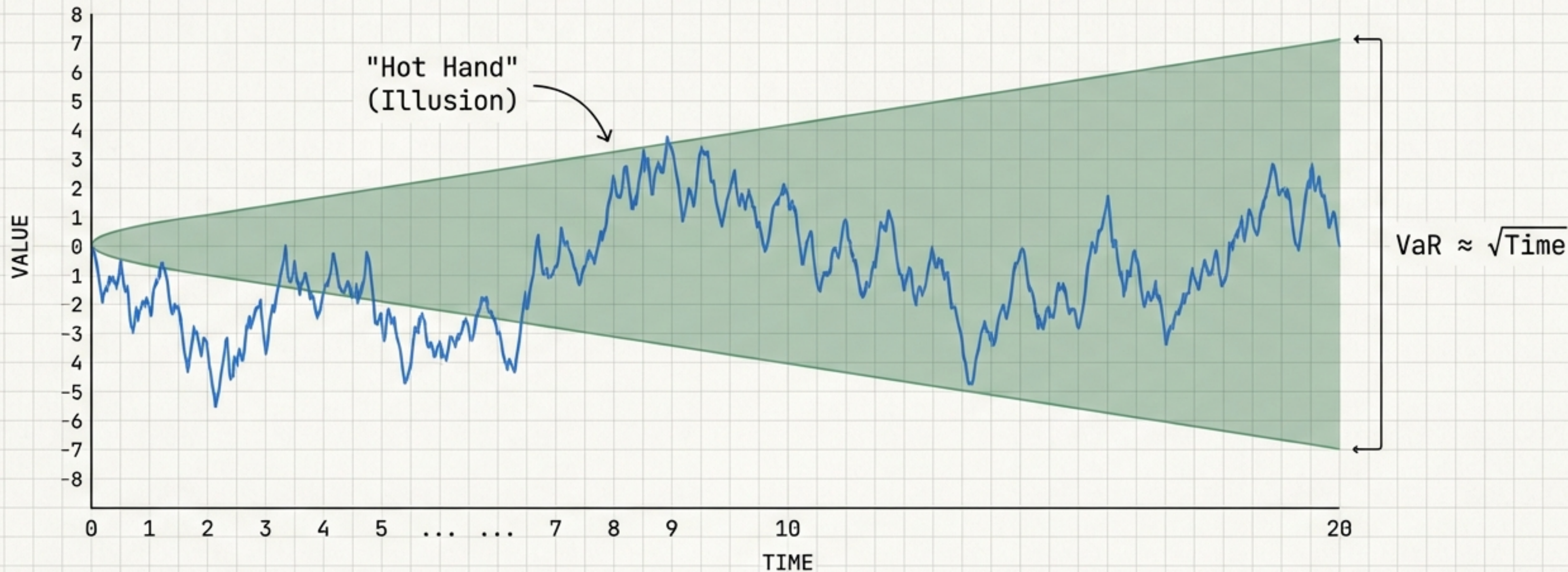
History Matters. Small, random events at the beginning of a process (drawing a red ball) can lock in a dominant outcome for decades. We are often stuck with 'Red' not because it's better, but because it was first.

Markov Models & Equilibrium



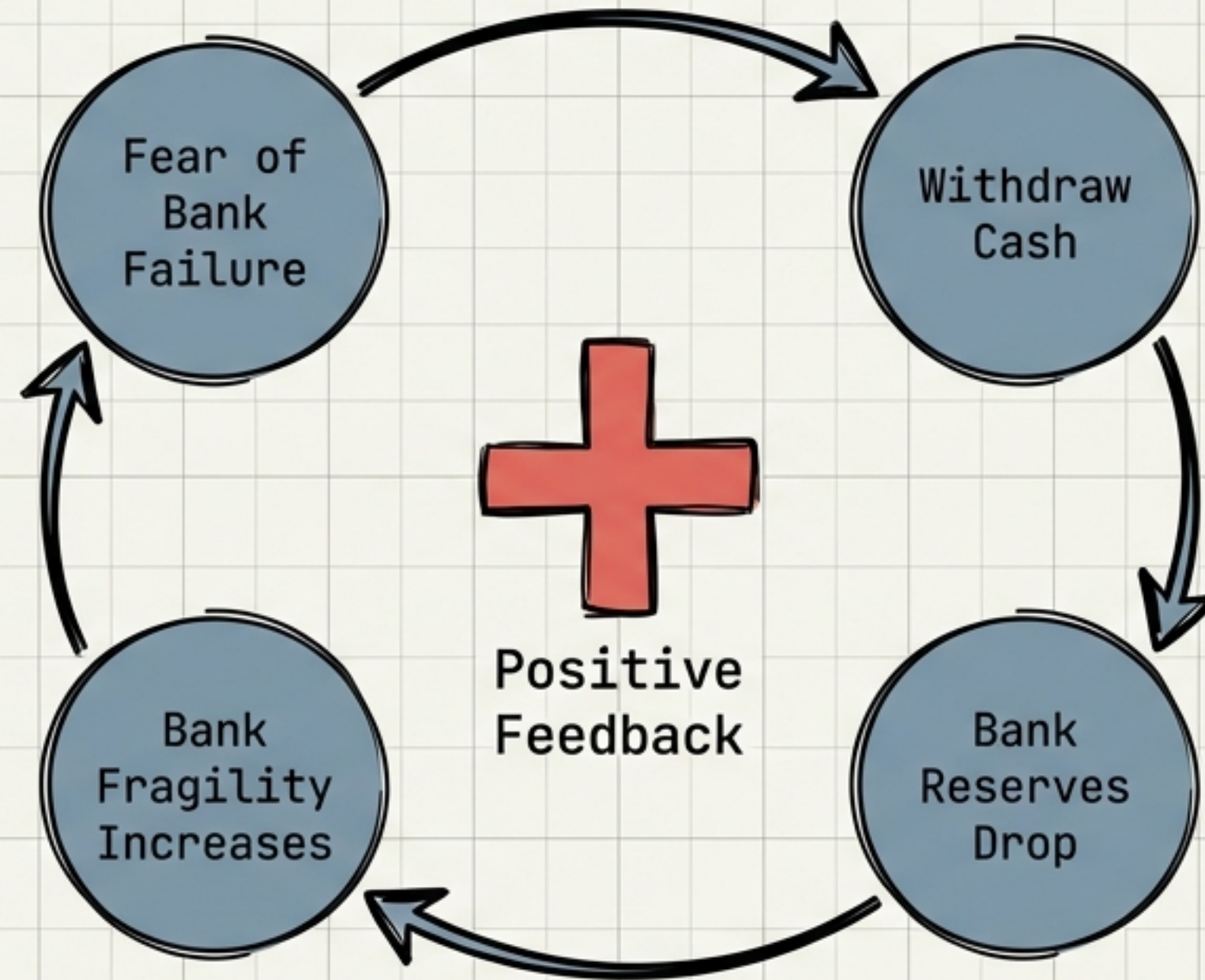
Trends are rarely linear lines to infinity. Markov models show us how systems circulate. Democracy isn't a one-way street; it's a dynamic equilibrium where countries transition back and forth, stabilizing at a mix.

Randomness vs. Pattern



We see patterns where there is only noise. A “Hot Hand” in sports or a stock streak is often just a Random Walk. Risk doesn’t grow linearly; it grows with the square root of time, which is why long-term predictions are so difficult.

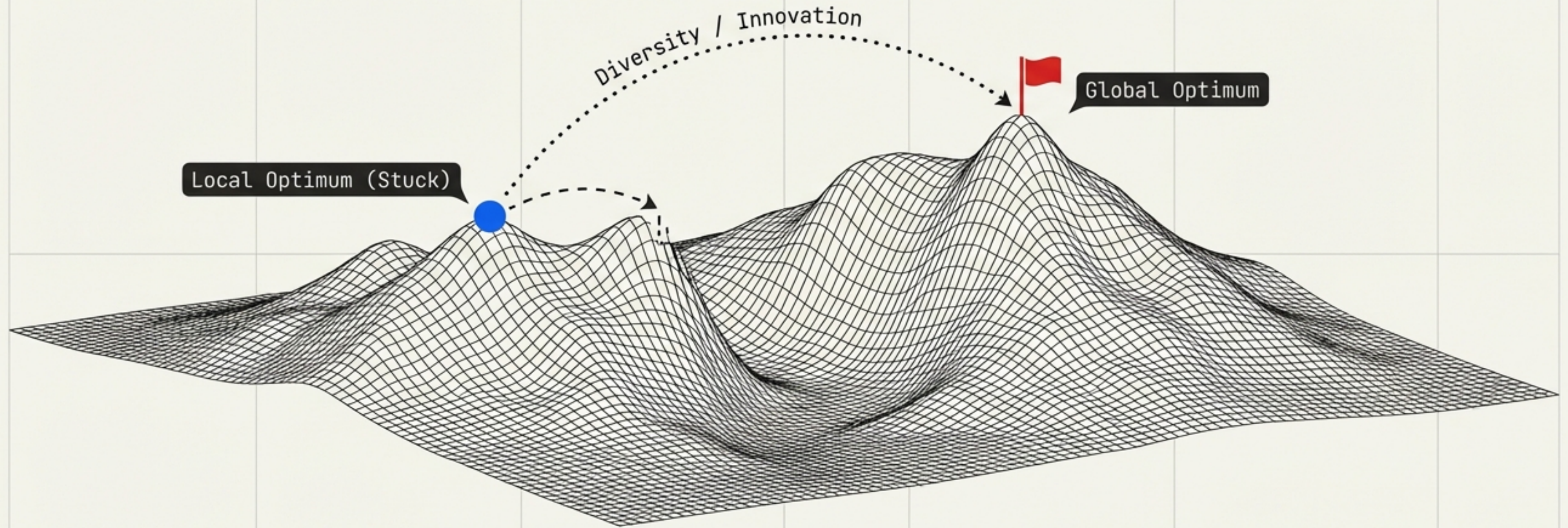
Feedback Loops: Vicious & Virtuous



Systems are defined by their loops. Positive feedback amplifies change, creating “Vicious Cycles” (Bank Runs) or “Virtuous Cycles” (Network Effects). To fix the system, you must break the loop.

MODULE E: INNOVATION

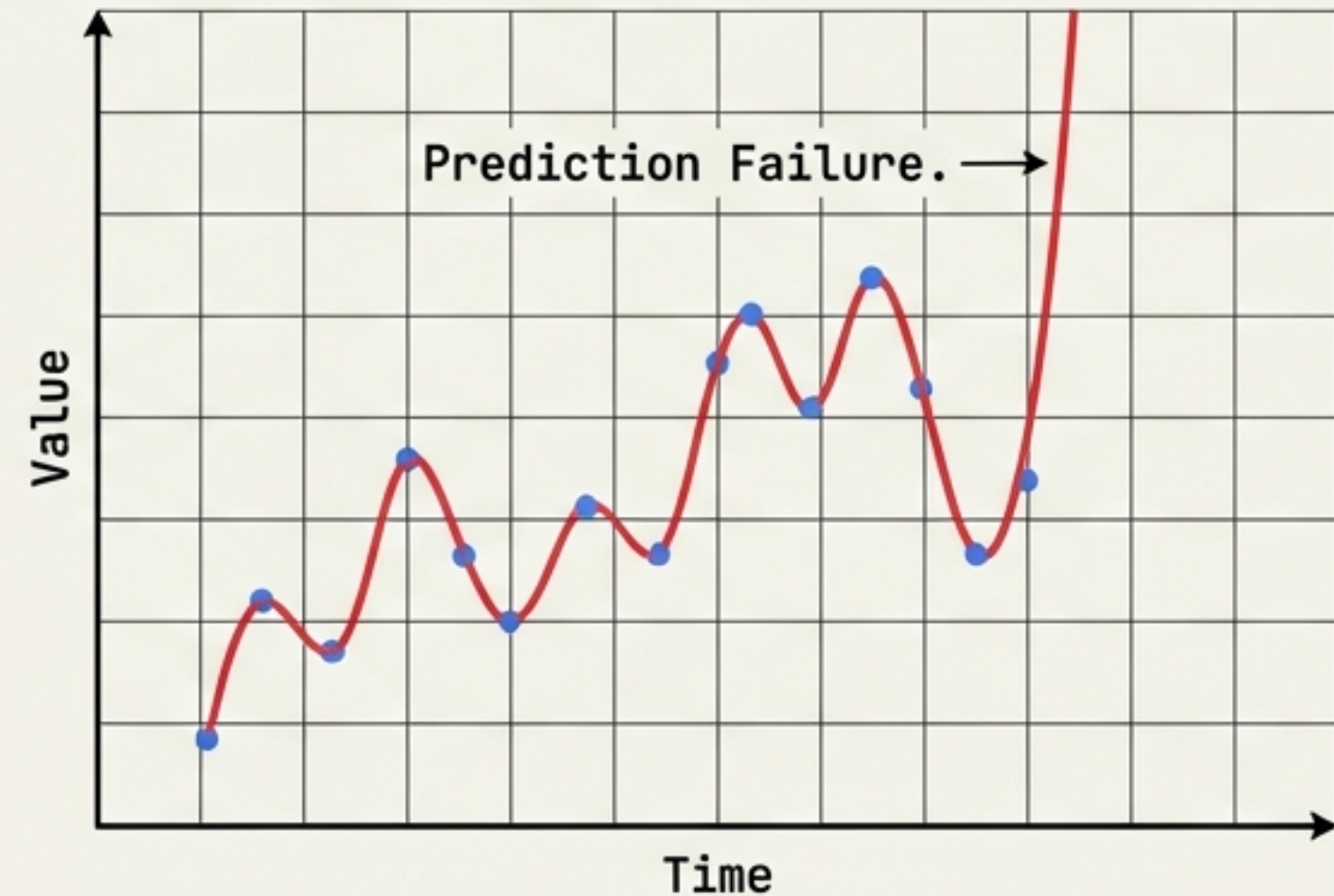
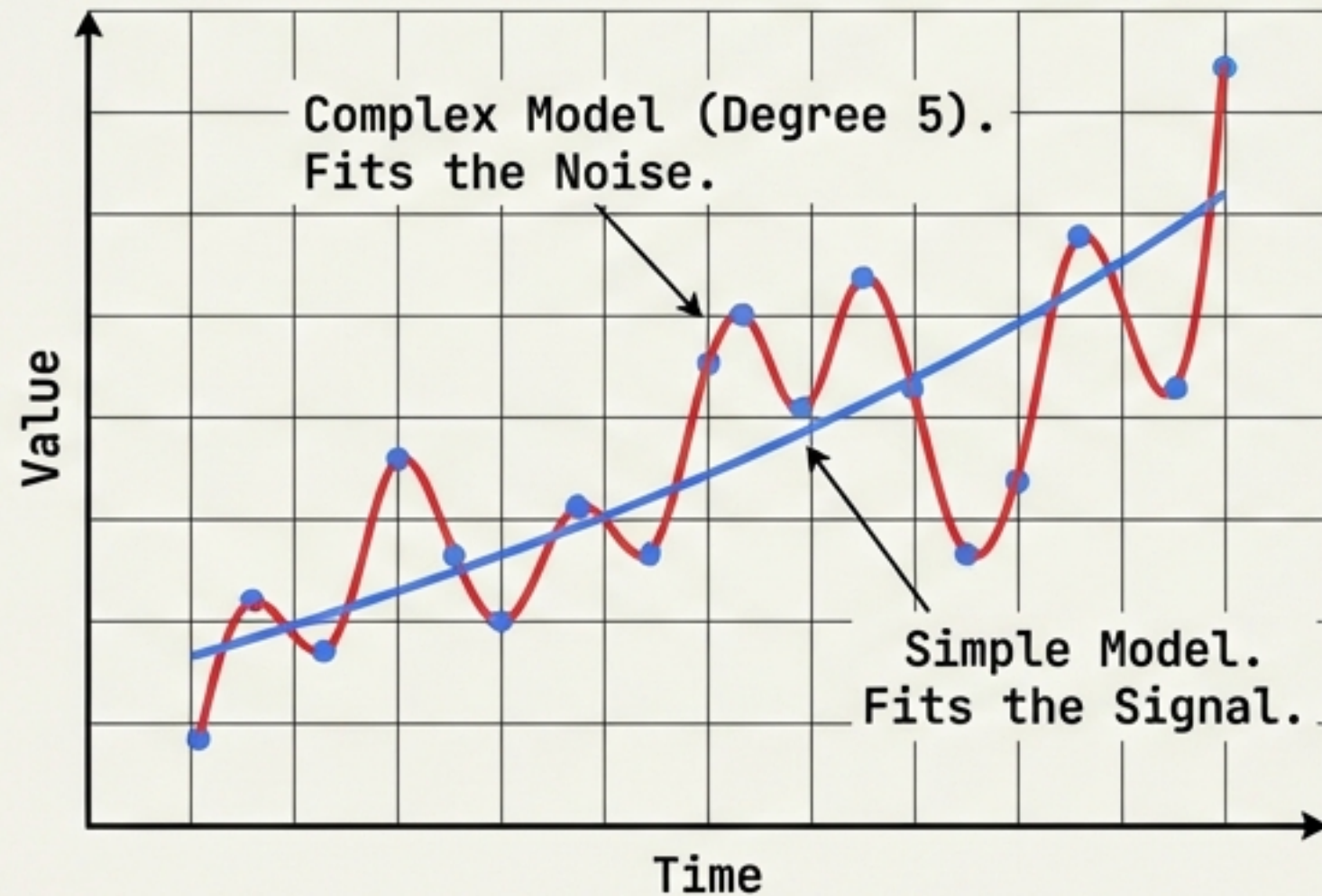
The Rugged Landscape



Optimization is like climbing a mountain in the fog. You climb until you can't go higher, but you might be stuck on a foothill (Local Peak). To find the highest peak (Global Optimum), you need to make non-incremental jumps—which requires diverse thinking.

The Trap of Overfitting

Overfitting and Out-of-Sample Error



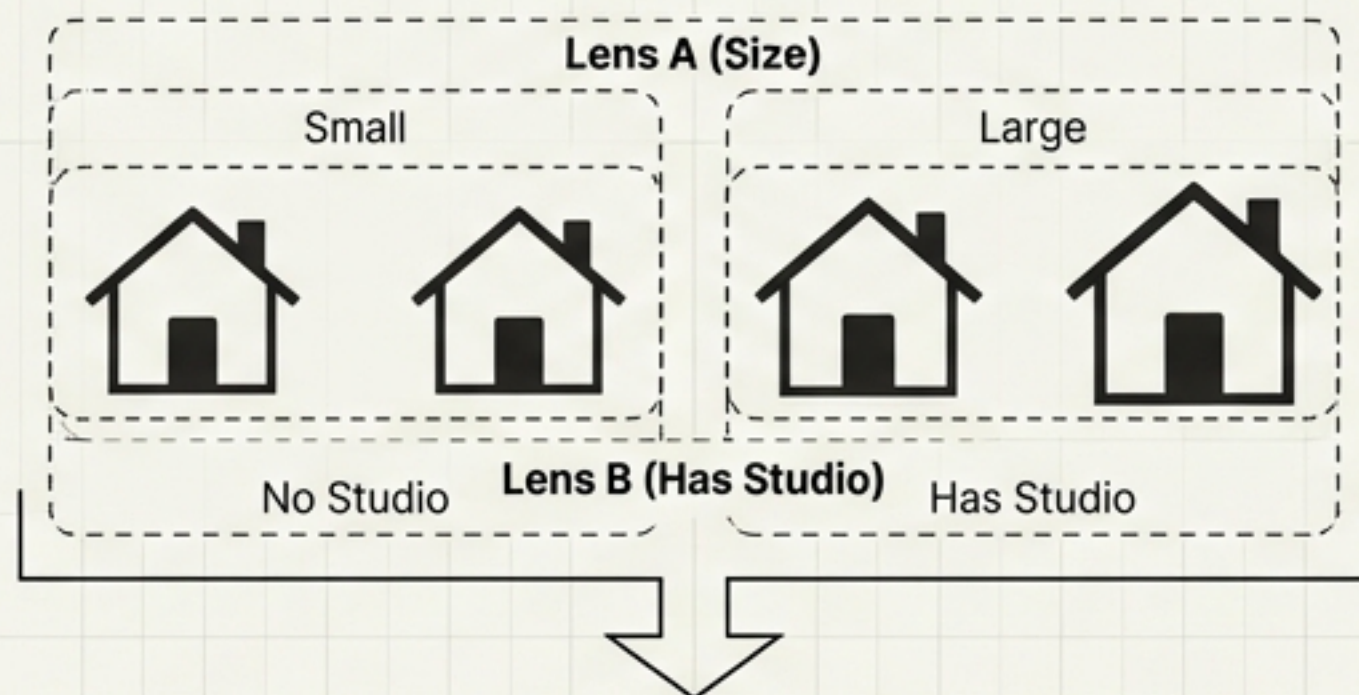
Complexity is not always accuracy. A model that is too complex “memorizes” the noise of the past and fails to predict the future. “Everything should be made as simple as possible, but not simpler.”

Architectural Precision

PHASE 3: WISDOM

The Diversity Prediction Theorem

Collective Error = Average Individual Error – Prediction Diversity



Why do many models beat one expert? The math proves it. To reduce the group's error, you can either make individuals smarter OR make them more different. Diversity isn't just nice; it's a mathematical component of accuracy.

Construct Your Lattice

PHASE 3: WISDOM



All models are wrong. Some are useful.

The world is too complex for a single lens. By building a lattice of models—Red Cape, Distributions, Systems, Games—we gain the geometry to Reason, Explain, Design, and Act.
This is the joy of the Model Thinker.

The practice of modeling can be a beautiful game. Through logical explorations, we improve and become wise, taking that wisdom to change the world in positive ways.